CHEMISTRY

1) For the cell reaction $2Ce^{+4} + Co \rightarrow 2Ce^{+3} + Co^{+2}$, $E°_{cell} = 1.89$ V and $E^{\circ}_{Co^{+2}/Co} = -0.28V$ then the value of $E^{\circ}_{Ce^{+4}/Ce^{+3}}$ will be :-

(1) –1.61 V
(2) +1.61 V
(3) +2.17 V
(4) +2.03 V

2) The measured voltage for the reaction with the indicated concentrations is 1.50 V. Calculate E^{o}_{cell}:-
$Cr(s) + 3Ag^{+}$ (aq, 0.10M) $\rightarrow 3Ag(s) + Cr^{3+}$(aq, 0.30M)

(1) 1.35 V
(2) 1.40 V
(3) 1.65 V
(4) 1.55 V

3) The time required for a current of 3 amp. to decompose electrolytically 18 g of H_2O is:

(1) 18 hour
(2) 36 hour
(3) 9 hour
(4) 18 seconds

4) Aluminium displaces hydrogen from acid solution but copper does not. A galvanic cell prepared by combining Cu/Cu^{2+} and Al/Al^{3+} has an emf of 2.0V at 298K. If the reduction potential of copper electrode is +0.34V, then reduction potential of aluminium electrode will be :-

(1) +1.66V
(2) –1.66V
(3) 2.34V
(4) –2.3V

5) Given: $E^{\circ}_{Fe^{3+}/Fe} = -0.036V$ $E^{\circ}_{Fe^{2+}/Fe} = -0.439V$. The value of standard electrode potential for the change, $Fe^{3+}(aq) + e^{-} \rightarrow Fe^{2+}(aq)$ will be :

(1) –0.072 V
(2) 0.385 V
(3) 0.770 V
(4) –0.270 V

6) In concentration cell, which is incorrect :-

(1) The electrode material and the solution in both half-cells are composed of the same substances

(2) Concentrations of solutions of the same substances is different in both half cells

(3) $E^{o}_{cell} = 0$

(4) The Nernst equation reduces to $E_{cell} = \frac{RT}{nF} \square nQ$

7) An electrochemical cell is represented as
$Pt_{(s)}|O_2(g)|H_3O^+(aq) \,||\, OH^-(aq)|O_2(g)|Pt(s)$
Half cell reaction of anode will be :-

(1) $2OH^- \rightarrow \frac{1}{2}O_2 + H_2O + 2e^-$

(2) $H_2O \rightarrow 2H^+ + \frac{1}{2}O_2 + 2e^-$

(3) $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$

(4) $O_2 \rightarrow O^{2+} + e^-$

8) The standard EMF for the cell reaction,
$Zn + Cu^{2+} \rightarrow Cu + Zn^{2+}$ is 1.10 V at 25°C. The EMF for the cell reaction, when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solutions are used, at 25°C is

(1) 1.10 V

(2) 0.110 V

(3) –1.10 V

(4) –0.110 V

9) Find SRP of M^{+2}/M electrode from following information :-
$2H^+(aq) + M(s) \rightleftharpoons M^{2+}(aq.) + H_2(g)$;
$\Delta G° = +96.5$ kJ at 298 K

(1) 0.5 volt

(2) –0.5 volt

(3) 1 volt

(4) –1 volt

10) Corrosion of iron is essentially an electrochemical phenomenon where the cell reactions are

(1) Fe is oxidised to Fe^{2+} and dissolved oxygen in water is reduced to O^{2-}

(2) Fe is oxidised to Fe^{3+} and H_2O is reduced to O_2^{2-}

(3) Fe is oxidised to Fe^{2+} and H_2O is reduced to O^{2-}

(4) Fe is oxidised to Fe^{2+} and H_2O is reduced to O_2

11) At 25°C the potential of hydrogen electrode (P_{H_2} = 1 atm and $[H^{\oplus}]$ = 0.04 M) will be :-

(1) +0.084 V

(2) –0.042 V

(3) –0.063 V

(4) –0.084 V

12) Based on the data given below, the correct order of reducing power is
Fe^{3+} (aq) + e^- → Fe^{2+} (aq) ; E° = 0.77 V
Al^{3+} (aq) + $3e^-$ → Al(s) ; E° = – 1.66 V
Br_2(aq) + $2e^-$ → $2Br^-$ (aq) ; E° = 1.08 V

(1) Br^- < Fe^{2+} < Al

(2) Fe^{2+} < Al < Br^-

(3) Al < Br^- < Fe^{2+}

(4) Al < Fe^{2+} < Br^-

13) One Faraday electricity is passed through the solutions of $AgNO_3$ and $CuSO_4$ connected in series. The mass of Ag and Cu deposited would be in the ratio :-
(Ag = 108, Cu = 63.5)

(1) 54 : 31.75

(2) 54 : 63.5

(3) 108 : 31.75

(4) 108 : 63.5

14) The equivalent conductance of Ba^{2+} and Cl^- are respectively 127 and 76 $ohm^{-1}cm^2eq^{-1}$ at infinite dilution. What will be the equivalent conductance of $BaCl_2$ at infinite dilution ?

(1) 139.5 $ohm^{-1}cm^2eq^{-1}$

(2) 203 $ohm^{-1}cm^2eq^{-1}$

(3) 279 $ohm^{-1}cm^2eq^{-1}$

(4) 101.5 $ohm^{-1}cm^2eq^{-1}$

15) Specific conductance of NaCl is maximum for which of the following solution :-

(1) 0.001 N

(2) 0.01 N

(3) 0.1 N

(4) 1.0 N

16) An electrolytic cell contains a solution of Ag_2SO_4 and has platinum electrodes. A current is passed untill 1.6 g of O_2 has been liberated at anode. The amount of silver deposited at cathode will be
(Molar mass of Ag = 108g, O_2 = 32g):-

(1) 1.08 g

(2) 1.6 g

(3) 0.8 g

(4) 21.60 g

17) $Cu^{+2} + 2e^{-} \rightarrow Cu$, on increasing concentration of Cu^{+2} electrode potential :-

(1) Increases

(2) Decreases

(3) No change

(4) First increases then decreases

18) One gram metal ion M^{+3} was discharged by the passage of 1.81×10^{23} electrons. What is the atomic weight of metal:-

(1) 33.35

(2) 133.4

(3) 66.7

(4) 10

19) If SOP of Cu/Cu^{+2} is –0.34 V, then the correct order of oxidation potential in volts of the below electrodes follow the order :
P = Cu^{+2} (0.0001 M) / $Cu_{(s)}$
Q = Cu^{+2} (0.1 M) / $Cu_{(s)}$
R = Cu^{+2} (0.01 M) / $Cu_{(s)}$
S = Cu^{+2} (0.001 M) / $Cu_{(s)}$

(1) Q > R > S > P

(2) P > S > R > Q

(3) R > S > Q > P

(4) P > Q > R > S

20) When a solution of $AgNO_3$(1M) is electrolyzed using platinum anode and copper cathode. What are the products obtained at two electrodes ? Given : $E^{o}_{Cu^{2+}|Cu} = +0.34\,volt$; $E^{o}_{O_2,\,H^+|H_2O} = +1.23\,volt$ $E^{o}_{H^+|H_2} = +0.0\,volt$; $E^{o}_{Ag^+|Ag} = +0.8\,volt$

(1) $Cu \rightarrow Cu^{2+}$ at anode ; $Ag^{+} \rightarrow Ag$ at cathode

(2) $H_2O \rightarrow O_2$ at anode ; $Cu^{2+} \rightarrow Cu$ at cathode

(3) $H_2O \rightarrow O_2$ at anode ; $Ag^{+} \rightarrow Ag$ at cathode

(4) $NO_3^{-} \rightarrow NO_2$ at anode ; $Ag^{+} \rightarrow Ag$ at cathode

21) Calculate cell potential at 298 K for following galvanic cell
$Cd(s)\,|Cd^{+2}(aq.)\,(0.1M)\,\|\,H^{+}_{(aq)}\,(0.1M)\,|H_2(g, 0.5\,atm)|Pt\;\; E^{\circ}_{Cd^{2+}/Cd} = -0.40V$

(1) 0.38 V

(2) – 0.38 V

(3) 0.36 V

(4) – 0.36 V

22) If hydrogen electrode dipped into solution of pH = 3 and pH = 6 and salt bridge is connected, the emf of resulting cell is :-

(1) 0.177 V

(2) 0.3 V

(3) 0.052 V

(4) 0.104 V

23) If standard reduction potentials of Ni^{+2}/Ni and Au^{+3}/Au half cells are –0.25 volt and 1.50 volt respectively, the E_{cell} of the cell,
Ni/Ni^{+2} (0.01M) || Au^{+3} (0.1 M) | Au will be

(1) greater than $E°_{cell}$

(2) Equal to $E°_{cell}$

(3) less than $E°_{cell}$

(4) Unpredictable

24) Standard electrode potential for Sn^{4+}/Sn^{2+} couple is +0.15V and that for the Cr^{3+}/Cr couple is –0.74V. These two couples in their standard state are connected to make a cell. The cell potential will be

(1) +1.19V

(2) +0.89V

(3) +0.18V

(4) +1.83V

25) 4.5 g of aluminium is deposited at cathode from Al^{3+} solution by certain quantity of electric charge. The volume of hydrogen gas produced at STP from H^+ ions in solution by the same quantity of electric charge will be :-

(1) 22.4 L

(2) 44.8 L

(3) 5.6 L

(4) 11.2 L

26)

A current of 9.65A is passed through 2L aqueous solution of NaCl using suitable electrodes for 1000s. The pH of solution after electrolysis will be :

(1) 12.7

(2) 8.3

(3) 5.7

(4) 2.7

27) A 1.0 M solution of Cd^{2+} is added to excess iron and the system is allowed to reach equilibrium. What is the concentration of Cd^{2+} at equilibrium ?
$Cd^{2+}(aq)+Fe(s) \rightleftharpoons Cd(s) + Fe^{2+}(aq)$; $E°=0.037V$

(1) 0.195 M

(2) 0.097 M

(3) 0.053 M

(4) 0.145 M

28) In fuel cell H_2 and O_2 react to produce electricity. In the process, H_2 gas is oxidized at anode and O_2 gas is reduced at cathode. If 67.2 litre of H_2 gas at STP produced in 15 min. then find out the current passed ?

(1) 64.33 A

(2) 6.433 A

(3) 643.33 A

(4) 38.6 A

29) From the following E° values, of half cells :
(i) $A + e \rightarrow A^-$, –0.34 V
(ii) $B^- + e \rightarrow B^{2-}$; E° = +1.30 V
(iii) $C^- + 2e \rightarrow C^{3-}$; E° = –1.30 V
(iv) $D + 2e \rightarrow D^{2-}$; E° = +0.78 V
What combination of two half cells would result in a cell with the largest standard potential ?

(1) (ii) and (iii)

(2) (ii) and (iv)

(3) (i) and (iii)

(4) (i) and (iv)

30) How long must a 50.0 A current be passed through an electrolytic cell for refining copper to produce 3.00 mol of copper ?

(1) 1.50 hrs

(2) 1.61 hrs

(3) 2.50 hrs

(4) 3.22 hrs

31) What will be the products obtained at anode and cathode respectively, on electrolysis of aqueous solution of $AgNO_3$ with platinum cathode and silver anode :-

(1) O_2, Ag

(2) $Ag^{\oplus}$, Ag

(3) $Ag^{\oplus}$, H_2

(4) Ag, $Ag^{\oplus}$

32) In acidic medium, MnO_4^- is converted to Mn^{2+}. The charge required to reduce 0.06 mole of MnO_4^-

would be :-

(1) 0.8 F
(2) 0.05 F
(3) 0.30 F
(4) 0.25 F

33) The metal that cannot be obtained by electrolysis of an aqueous solution of its salt is :

(1) Cu
(2) Cr
(3) Ca
(4) Ag

34) Ratio of moles of Ag^{+} ion, Mg^{+2} ion, Al^{+3} ion would be deposited by passage of 3F charge in its solution :-

(1) $1 : \frac{1}{2} : \frac{1}{3}$
(2) $\frac{1}{3} : \frac{1}{2} : 1$
(3) 1 : 2 : 3
(4) 1 : 1 : 1

35) Standard oxidation potentials of four metals P, Q, R and S are +2.87, +3.05, –0.80 and +0.25 volt respectively. The reducing power of these metals are :-

(1) P > Q > R > S
(2) S > R > Q > P
(3) Q > P > S > R
(4) None of these

36) Specific conductance of 0.1 M Nitric acid is 6.3×10^{-2} ohm^{-1} cm^{-1}. The molar conductance of the solution is :

(1) 630 ohm^{-1} cm^{2} mol^{-1}
(2) 315 ohm^{-1} cm^{2} mol^{-1}
(3) 100 ohm^{-1} cm^{2} mol^{-1}
(4) 6.300 ohm^{-1} cm^{2} mol^{-1}

37) Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1 M is 100 Ω. The conductivity of this solution is 1.29 S m^{-1}. Resistance of the same cell when filled with 0.02 M of the same solution is 520 Ω. The molar conductivity of 0.02 M solution of the electrolyte will be.

(1) 124×10^{-4} S m^{2} mol^{-1}

(2) 1240×10^{-4} S m^2 mol^{-1}

(3) 1.24×10^{-4} S m^2 mol^{-1}

(4) 12.4×10^{-4} S m^2 mol^{-1}

38) Resistance of 0.1 M KCl solution in a conductance cell is 300 ohm and conductivity is 0.013 Scm^{-1}. The value of cell constant is –

(1) 3.9 cm^{-1}

(2) 39 m^{-1}

(3) 3.9 m^{-1}

(4) None of these

39) Equivalent conductivity of $Fe_2(SO_4)_3$ is related to molar conductivity by the expression :-

(1) $\Lambda_{eq} = \Lambda_m$

(2) $\Lambda_{eq} = \frac{\Lambda_m}{3}$

(3) $\Lambda_{eq} = 3\Lambda_m$

(4) $\Lambda_{eq} = \frac{\Lambda_m}{6}$

40) Select the incorrect statement among the following.

(1) The SI unit of conductivity is Sm^{-1}

(2) E_{cell} is an intensive parameter

(3) A positive E° means that the redox couple is stronger reducing agent than H^+/H_2 couple

(4) Magnitude of conductivity depends on the nature of the material

41) If x is specific resistance (in S^{-1} cm) of the electrolyte solution and y is the molarity of the solution, then Λ_m (in S cm^2mol^{-1}) is given by :-

(1) $\frac{1000x}{y}$

(2) $\frac{y}{1000x}$

(3) $\frac{1000}{xy}$

(4) $\frac{xy}{1000}$

42) Equivalent conductance of saturated $BaSO_4$ is 400 ohm^{-1} cm^2 $equi^{-1}$ and specific conductance is 8×10^{-5} ohm^{-1} cm^{-1}. Hence, K_{sp} of $BaSO_4$ is.

(1) 4×10^{-8} M^2

(2) 1×10^{-8} M^2

(3) 2×10^{-4} M^2

(4) $1 \times 10^{-4}\ M^2$

43) The resistance of 0.1 N solution of formic acid is 200 ohm and cell constant is 2.0 cm^{-1}. The equivalent conductivity (in Scm^2eq^{-1}) of 0.1 N formic acid is :-

(1) 100

(2) 10

(3) 1

(4) None of these

44) Match the following process with quantity of charge required

Column-I		**Column-II**	
(A)	Electrolysis of H_2O to give 5.6L O_2 at STP	(P)	3F
(B)	Conversion of 0.5 mole $Cr_2O_7^{2-}$ into Cr^{3+}	(Q)	4F
(C)	Reduction of 18g Al^{3+} into Al	(R)	1F
(D)	Electrolysis of 2L, 1M $CuSO_4$ solution to give Cu	(S)	2F

(1) A → Q, B → P, C → S, D → R

(2) A → S, B → R, C → Q, D → P

(3) A → R, B → P, C → S, D → Q

(4) A → P, B → S, C → R, D → Q

45) Calculate emf of the cell in which the following reaction takes place :-
$Ni_{(s)} + 2Ag^+(0.002\ M) \rightarrow Ni^{+2}(0.16\ M) + 2Ag_{(s)}$
Given : $E^{\circ}_{cell} = 1.05$ V

(1) 1.05 V

(2) 0.912 V

(3) 1.19 V

(4) 2.05 V

46) ***Assertion :-*** Molar conductivity of an electrolytic solution increases on dilution.
Reason :- Decrease in conductivity is caused by increase in volume.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

47) **Assertion (A) :** The electrode potential of SHE is zero only at 25°C and not at any other temperature.

Reason (R) : SHE is reference electrode.

(1) Both assertion and reason are true but reason is not correct explanation of assertion.

(2) Assertion is true but reason is false

(3) Assertion is false but reason is true

(4) Both assertion and reason are true and reason is a correct explanation of assertion.

48) **Assertion (A) :** If an aqueous solution of NaCl is electrolysed, the product obtained at the cathode is H_2 gas and not Na metal.
Reason (R) : Gases are liberated faster than the metals during the electrolysis of an electrolyte.

(1) Both assertion and reason are true but reason is not correct explanation of assertion

(2) Assertion is true but reason is false

(3) Assertion is false but reason is true

(4) Both assertion and reason are true and reason is a correct explanation of assertion

49) Match List-I with List-II :-

	List-I		**List-II**
(A)	$2H_2(g) + O_2(g) \rightarrow 2H_2O(\square)$	(I)	Dry cell
(B)	Cell potential remains constant	(II)	Corrosion
(C)	Development of green coating on copper	(III)	Fuel cell
(D)	Cell used commonly in transistors and clocks	(IV)	Mercury cell

Choose the correct answer from the options given below :-

(1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(3) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

50) Match the terms given in Column-I with their units in Column-II.

	Column-I		**Column-II**
(A)	Molar conductivity	(P)	Sm^{-1}
(B)	Conductance	(Q)	Sm^2mol^{-1}
(C)	Conductivity	(R)	m^{-1}
(D)	Cell constant	(S)	$ohm^{-1}(S)$

(1) A–Q, B–P, C–S, D–R

(2) A–Q, B–S, C–P, D–R

(3) A–S, B–Q, C–R, D–P

(4) A–P, B–S, C–Q, D–R

BIOLOGY

1) Read the following terminology of column-I & column-II and identify their correct match : -

Column-I		Column-II	
(A)	Testes	(i)	Fertilization of gametes
(B)	Ovaries	(ii)	Sperm formation
(C)	Fallopian tube	(iii)	Help in Parturition
(D)	Uterus	(iv)	Ovum formation
		(v)	Development of foetus
		(vi)	Help in digestion
		(vii)	Milk formation

(1) A-ii, B-i, C-iii, D-iv

(2) A-i, B-ii, C-iii, D-vii

(3) A-ii, B-iv, C-i, D-v

(4) A-iii, B-vii, C-ii, D-i

2) Read column I, II & III and identify their correct match : -

Column-I		Column-II			Column-III
(A)	LNG-20	(P)	Oral pills	(i)	Reduce sperm motility
(B)	Mala-D	(Q)	Hormonal IUD	(ii)	prevent entry of sperm
(C)	Multiload 375	(R)	Barrier	(iii)	Prevent ovulation
(D)	Vault	(S)	Copper releasing IUD	(iv)	Cervix hostile to the sperm

(1) A-P- iv, B-Q - i, C - R-ii, D-S - iii

(2) A-R-i, B-S - ii, C - P-iii, D-Q - iv

(3) A-S- iii, B-P - ii, C - Q-i, D-R - iv

(4) A-Q- iv, B-P - iii, C - S-i, D-R - ii

3) Find out incorrect pair :-

(1) Formation of gametes - primary sex organ

(2) Fertilization - fallopian tube

(3) Insemination - Male genital tract

(4) Embryonic development- uterus.

4) How many of the following male accessary glands are paired.
(a) seminal vesicle (b) prostate
(c) bulbourethral (d) bartholin

(1) Two
(2) One
(3) Four
(4) Three

5) Secretions of seminal vesicle, prostate and bulbourethral glands constitute the ___, Which is rich in fructose, calcium and certain enzymes.

(1) Seminal fluid
(2) Prostatic fluid
(3) bulbourethral fluid
(4) Seminal plasma

6) **Statement-I : -** Glandular tissue of each breast is divided into 15-20 mammary lobes.
Statement-II : - Several mammary ducts join to form a wider mammary ampulla.

(1) Statement I is correct but Statement II is incorrect.
(2) Statement-I is incorrect & statement-II is correct.
(3) Both statements are incorrect.
(4) Both statements are correct.

7) **Statement-(1) : -** The sperm head contains an elongated haploid nucleus: -
Statement-(2) : - The acrosome filled with mitochondria that help in fertilization of ovum.

(1) Both statement-1 & statement-2 are correct.
(2) Both statement-1 & statment-2 are incorrect.
(3) Statement-1 is correct & statement-2 is incorrect.
(4) Statement-1 is incorrect & statement-2 is correct.

8) **Statement-1 : -** No more oogonia are formed and added after birth.
Statement-2 : - Oogenesis is initiated during puberty.

(1) Statement-1 is correct & statement-2 is incorrect
(2) Statement-1 is incorrect & statement-2 is correct
(3) Both statement 1 & 2 are correct.
(4) Both statement 1 & 2 are incorrect.

9) Statement-1 : Tertiary follicle is characterised by a fluid filled cavity called antrum.
Statement-2 : Release of secondary oocyte from the ovary by the process called ovulation.

(1) Both statement-1 & 2 are incorrect.
(2) Both statement-1 & 2 are correct.
(3) Statement-1 is correct & statement-2 is incorrect.

(4) Statement1 is incorrect & statement-2 is correct.

10) **Statement-1 : -** Oocyte /ovum along with egg membranes is termed as egg.
Statement-2 : - Perivitelline space is the space between plasma membrane of oocyte & zona pellucida.

(1) Both statement 1 & 2 are incorrect.
(2) Both statements 1 & 2 are correct
(3) Statement-1 is correct & statement-2 is incorrect
(4) Statement-1 is incorrect & statement-2 is correct

11) Fill in the blanks A, B and C respectively : -
Human egg has __''A''__ egg membranes from inside to outside that is __''B''__ and __''C''__ .

(1) A-Three , B - Zona pellucida C- Corona radiata
(2) A-Two, B - Corona radiata C- Zona pellucida
(3) A-Three, B - Corona radiata C- Zona pellucida
(4) A-Two, B - Zona pellucida C- Corona radiata

12) Select the incorrect pair :-

(1) Cleavage - Repeated mitotic division
(2) New cell form during cleavage - Blastomere
(3) Cleavage starts - Isthmus of fallopian tube
(4) 8-16 cell stage - Blastocyst

13) **Statement-1 : -** According to world health organization, reproductive health means a total well being in all aspects of reproduction.
Statement-2 : - Family planning programmes were initiated in 1951.

(1) Statement-1 is correct & statement-2 is incorrect
(2) Statement-2 is correct & statement-1 is incorrect
(3) Both statement 1 & 2 are incorrect.
(4) Both statement 1 & 2 are correct.

14) The world population was in 2011 : -

(1) 2 billion
(2) 6 billion
(3) 7.2 billion
(4) 1.2 billion

15) Find out incorrect pair : -

(1) After one month of pregnancy - Embryo's heart is formed
(2) By the end of second month of pregnancy - foetus develops limbs & digits
(3) By the end of third months of pregnancy - first movement of foetus

(4) By the end of nine month of pregnancy - foetus is fully developed .

16) Which of the following hormones are not secreted by placenta during pregnancy ?

(1) Estrogen, HCG
(2) FSH, Inhibin
(3) Progesterone, HCG
(4) HCG, hPL

17) Sperm entry into the ovum induces the completion of the meiotic division of the

(1) Primary oocyte
(2) Secondary oocyte
(3) Ootid
(4) Mature ovum

18) Which of following organ in female is responsible for embryo development and child care respectively?

(1) Uterus, fallopion tube
(2) Uterus, Mammary gland
(3) Ovary, Bartholin's gland
(4) Mammary gland, uterus

19) Lack of menstruation may be due to the :-

(1) Stress
(2) Poor health
(3) Pregnancy
(4) All of the above

20) The process by which spermatids are transformed into spermatozoa called-

(1) Spermatogenesis
(2) Spermiation
(3) Spermiogenesis
(4) Capacitation

21) Cilia present in fallopian tube helps in movement of :

(1) Ovum
(2) Zygote
(3) Early embryo
(4) All of these

22) In human the testes are situated outside the abdominal cavity with in a pouch called -

(1) Scrotum

(2) Epididymis

(3) Fore skin

(4) Glans penis

23) How many of the following reproductive events occur in humans :
(A) Gametogenesis (B) Insemination
(C) Fertilization (D) Implantation
(E) Gestation (F) Parturition

(1) Two

(2) Four

(3) Five

(4) Six

24) Which one of the following is incorrectly paired with its function ?

(1) Ovary - Secretion of feminine hormone

(2) Fallopian tube - Collection of ova

(3) Seminal vesicle - Nourishment to sperm

(4) Bartholin glands - Neutralise the acidity of urethra

25) Identify the correct statement about "inhibin" :-

(1) Inhibits the secretion of LH, FSH and prolactin.

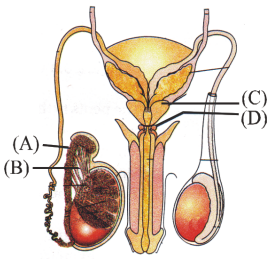
(2) Inhibin produced by granulosa cells of ovary and inhibits the secretion of FSH.

(3) Inhibits the secretion of testosterone.

(4) Is produced by nurse cells and inhibits the secretion of LH only.

26)

Identify A, B, C and D in the given diagram. Which of the following among these has been correctlly identified along with its function-



(1) (A) Epididymis - Formation of sperm

(2) (B) Rete testis - Functional maturation of sperms

(3) (C) Seminal vesicle - store and transport the sperms

(4) (D) Bulbourethral gland - helps in the lubrication of penis

27) Match the Column-I and Column-II and Choose the correct option.

	Column I		Column II
(a)	Mons pubis	(1)	Fleshy folds of tissue
(b)	Labia majora	(2)	Cushion of fatty tissue
(c)	Hymen	(3)	Tiny fingerlike structure
(d)	Clitoris	(4)	Partial covers the vaginal orifice

Option :
(1) (a)–2, (b)–1, (c)–4, (d)–3
(2) (a)–4, (b)–3, (c)–2, (d)–1
(3) (a)–1, (b)–4, (c)–3, (d)–2
(4) (a)–2, (b)–1, (c)–3, (d)–4

28) ***Assertion (A) :-*** Colostrum from mother is essential for new born babies.
Reason (R) :- Colostrum contains antibodies.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

29) Identify the **incorrect** pair :-

(1) Tertiary follicle → Antrum
(2) Ovulation → Due to LH hormone
(3) Spermatids transformed into → Spermiogenesis spermatozoa
(4) Capacitation → occurs in epididymis

30)

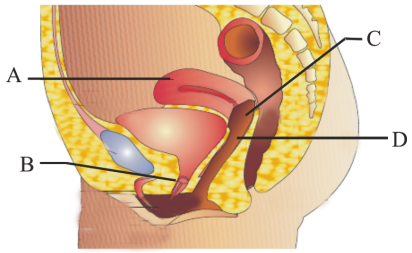
Endocrine cells, found outside seminiferous tubules are :-

(1) Sertoli cells
(2) Leydig cells
(3) Spermatogonia
(4) Immunocompetant cells

31) The cellular layer that disintegrates and regenerates again and again in human in menstrual cycle :-

(1) Endothelium of blood vessels
(2) Germinal epithelium of ovary
(3) Tunica propria of seminiferous tubules
(4) Endometrium of uterus

32) It is a diagrammatic sectional view of female reproductive system. In which identify the site



where the embryo develop?

(1) A
(2) B
(3) C
(4) D

33) Which sequence is correct of male genital tract?

(1) Rete testis → Vas deferens → Vasa efferentia → Epididymis → Urethra
(2) Tubuli recti → Rete testis → Vas deferens → Vasa efferenta → Seminal vesicle → Urethra
(3) Tubuli recti → Rete testis → Vasa efferentia → Vas deferens → Epididymis → Urethra
(4) Tubuli recti → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Urethra

34) When released from ovary, normally human egg contains:-

(1) 1 Y chromosome
(2) 2 X chromosome
(3) 1 X chromosome
(4) XY chromosome

35) Read the following statements carefully.
The .....(A)..... present on the inside wall of .....(B)..... multiply by .....(C)..... division. A .....(D)..... completes the .....(E)..... division leading to formation of two equal, haploid cells called .....(F)..... . Which one of the following options gives the correct filling of A, B, C, D, E and F?

	(1)	(2)	(3)	(4)
(A)	Spermatogonia	Primary Spermatocyte	Spermatogonia	Primary Spermatocyte
(B)	Seminiferous tubule	Sertoli Cell	Sertoli Cell	Seminiferous tubule
(C)	Mitotic	Mitotic	First meiotic	First meiotic
(D)	Primary Spermatocyte	Secondary Spermatocyte	Secondary Spermatocyte	Spermatid
(E)	First meiotic	First meiotic	Second meiotic	Second meiotic
(F)	Secondary Spermatocyte	Spermatid	Spermatid	Sperm

(1) 1

(2) 2

(3) 3

(4) 4

36) Read the following statements and choose the correct option
Statement A : Oxytocin secretion results in strong uterine contractions.
Statement B : The mammary glands undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy by the process called lactation.

(1) Both the statements are incorrect

(2) Both the statements are correct

(3) Only statement A is correct

(4) Only statement B is correct

37) Read the following Assertion (A) and Reason (R) and choose the correct option
Assertion (A): During proliferative phase, endometrium of uterus regenerates through proliferation.
Reason (R): During this phase, the primary follicles in the ovary grow to become mature follicle and these developing follicles secrete estrogen.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

38) Cleavage is a A which starts just after ferlilization as zygote move through B of the oviduct towards the uterus.

(1) A = Mitosis, B = Isthmus

(2) A = Meiosis, B = Isthmus

(3) A = Mitosis, B = Ampulla

(4) A = Amitosis, b = Ampulla

39) Statement-I :- Thin epithelium encloses the stroma part of ovary.
Statement -II :- Stroma is divided into a peripheral medulla and a inner cortex.

(1) Statement I and II both false.

(2) Statement I and II both true.

(3) Statement I is true but II is false.

(4) Statement I is false but II is true.

40) For normal fertility, atleast A sperms must have normal shape and size & atleast B of them must show vigorous motility. Identify the A and B in above statement.

(1) A = 40 %, B = 60%

(2) A = 60%, B = 40%

(3) A = 40%, B = 100 %

(4) A = 60%, B = 20%

41) Select the hormone which is not correctly matched with its function :
(A) Estrogen – Endometrial hyperplasia
(B) hCG – maintains corpus luteum.
(C) Oxytocin – strong uterine contractions
(D) Prolactin – milk synthesis
(E) Progesterone – promote foetal ejection reflex

(1) A and E

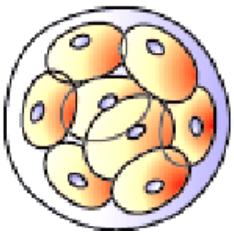
(2) Only D

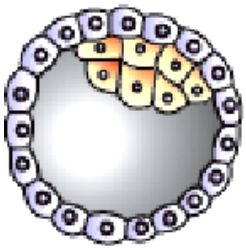
(3) D and E

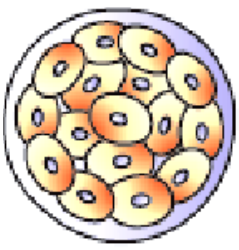
(4) Only E

42) Among the following identify the structure which is involved in implantation:-

(1) 

(2) 

(3) 

(4) 

43) Read following terminology of column I & II & identify their correct match ?

	Column I		**Column II**
(A)	Placenta	(i)	Inner cell mass

(B)	Blastomere	(ii)	Non-steroidal pill
(C)	Saheli	(iii)	Oral contraceptive pill
(D)	LNG-20	(iv)	Temporary endocrine gland
		(v)	Hormone releasing IUDs
		(vi)	Trophoblast

(1) A – iv, B – i, C – ii, D – v

(2) A – iv, B – i, C – ii, iii, D – iv

(3) A – i, B – ii, vi, C – iii, D – v

(4) A – vi, B – iv, C – ii, D – iii

44) How many of the given cells are haploid during gametogenesis-
Spermatogonium, Spermatid, Spermatozoa, Secondary spermatocyte, Secondary oocyte, Primary oocyte, oogonia

(1) 4

(2) 3

(3) 2

(4) 5

45) Match the Column I with Column II and choose the correct option

Column-I		**Column-II**	
(A)	ZIFT	(i)	Semen is artificially introduced into the uterus
(B)	IUT	(ii)	The zygote could be transferred into the fallopian tube
(C)	IUI	(iii)	Embryo with more than 8 blastomeres could be transferred into the uterus
(D)	ICSI	(iv)	Sperm is directly injected into the ovum

	(A)	(B)	(C)	(D)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iii)	(i)	(iv)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(i)	(iii)	(iv)	(iii)

(1) 1

(2) 2

(3) 3

(4) 4

46) Select the incorrect statement :

(1) Amniocentesis is ban for sex determination

(2) This procedure is used to test for the presence of certain genetic disorders.

(3) In Amniocentesis fluid taken from embryo.

(4) This procedure is performed most often between 14th and 16th week of pregnancy.

47) Read following Column-I & Column-II and identify their correct match :-

	Column-I		**Column-II**
A.	Non mediated IUD	i.	Non Hormonal pills
B.	Saheli	ii.	Lippes loops
C.	Hormonal IUD	iii.	LNG-20
D.	MALA-N	iv.	Cu-T
		v.	Hormonal oral pills
		vi.	Hormonal Injection

(1) (A–i), (B–iii), (C–iv), (D–vi)

(2) (A–iii), (B–iv), (C–v), (D–i)

(3) (A–ii), (B–i), (C–iii), (D–v)

(4) (A–v), (B–ii), (C–iv), (D–ii)

48) Medical assistance and care to people is required in reproduction related problems like
(A) Pregnancy (B) Delivery
(C) STDs (D) Abortions
(E) Contraceptions
(F) Menstrual problems
(G) Infertility

(1) A, B, C, D

(2) A, E, G

(3) B, C, D, F

(4) A, B, C, D, E, F & G

49) Which one of the following option gives correct description about these two barrier methods of contraception?

	Condoms	**Diaphragms**
1	Available for males only	Available for females only
2	They are disposable	They are reusable
3	Prevents insemination	Prevents ovulation
4	Covers penis	Placed inside uterine cavity

(1) 1

(2) 2

(3) 3

(4) 4

50) Which of the following is protozoan STD.

(1) Syphilis
(2) Gonorrhoea
(3) Trichomoniasis
(4) Genital herpes

ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	1	2	3	4	2	1	1	1	4	1	3	2	4	4	1	4	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	1	2	3	1	3	3	1	4	4	3	3	1	3	1	1	1	4	3
Q.	41	42	43	44	45	46	47	48	49	50										
A.	3	2	1	3	2	1	3	2	3	2										

BIOLOGY

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	3	4	3	1	4	4	3	1	2	2	4	4	4	3	3	2	2	2	4	3
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	1	4	4	2	4	1	4	4	2	4	1	4	3	1	2	1	1	3	2
Q.	91	92	93	94	95	96	97	98	99	100										
A.	4	3	1	1	2	3	3	4	2	3										

SOLUTIONS

CHEMISTRY

4) $E^{o}_{cell} = E^{o}_{Cu^{+2}/Cu} - E^{o}_{Al^{+3}/Al}$

$\Rightarrow$ 2.0 = (+0.34) – $E^{o}_{Al^{+3}/Al}$

$\Rightarrow$ $E^{o}_{Al^{+3}/Al}$ = –1.66 V

6) $E_{cell} = -\left(\frac{0.0591}{n}\right)$ log Q at 25°C

7) At anode O_2 is being liberated

$\Rightarrow H_2O \longrightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-$

9)

Class-XII, Part-I, Electrochemistry, (Page - 74) (2021-22 edition)

11)

$2H^+ + 2e^- \rightarrow H_2$

$$E_{H^+H_2} = 0 - \frac{0.0591}{2} \log \frac{P_{H_2}}{[H^+]^2}$$

$$E_{H^+H_2} = \frac{-0.0591}{2} \log \frac{1}{\left(4 \times 10^{-2}\right)^2}$$

= – 0.084 V

13) $\frac{W_1}{E_1} = \frac{W_2}{E_2} \Rightarrow \frac{W_1}{W_2} = \frac{108}{31.5}$

16) $\left(\frac{w}{E}\right)_{O_2} = \left(\frac{w}{E}\right)_{Ag}$

$\frac{1.6}{8} = \frac{w}{108}$; w = 21.6 g

18) Eq. wt. is liberated by 96500 coulomb i.e. 1 mole of electron (6.02×10^{23} electrons)
At. wt. = Eq. wt × 3

19)

$Cu_{(s)} \rightarrow Cu^{+2}_{(aq)} + 2e^-$

$Q = [Cu^{+2}]$

$E = E^0 - \frac{0.059}{2} \times \log[Cu^{+2}]$

$[Cu^{+2}]\uparrow, E\downarrow$

23) $2Au^{+3} + 3Ni \rightarrow 3Ni^{+2} + 2Au$

$$Q = \frac{[Ni^{+2}]^3}{[Au^{+3}]^2} = \frac{(0.01)^3}{(0.7)^2} = \frac{10^{-6}}{10^{-2}} = 10^{-4}$$

$$E_{cell} = E^o_{cell} - \frac{0.059}{6} \log 10^{-4}$$

$$E_{cell} = E^o_{cell} + \frac{4}{6} \times 0.059 \log 10$$

$\therefore \quad E_{cell} > E°_{cell}.$

28)

$$W = \frac{E}{F} \times i \times t$$

$$\therefore \ i = \frac{67.2}{22.4} \times 2 \times \frac{96500}{1 \times 900} = 643.33 \text{ amp.}$$

35)

Reducing power $\propto$ SOP

45) $Ni_{(s)} + 2Ag^+ \longrightarrow Ni^{+2} + 2Ag_{(s)}$

$$Q = \frac{[Ni^{+2}]}{[Ag^+]^2} = \frac{0.16}{(0.002)^2} = \frac{0.16}{(2 \times 10^{-3})^2} = \frac{0.16}{4 \times 10^{-6}}$$

$$Q = \frac{0.16 \times 10^6}{4} = \frac{16 \times 10^4}{4} = 4 \times 10^4$$

$E_{cell} = E^o_{cell} - \frac{0.059}{n} \log Q$ (From Nernst equation)

$E_{cell} = 1.05 - \frac{0.06}{2} \log(4 \times 10^4)$

$= 1.05 - 0.03\{\log 4 + \log_{10} 4\}$

$= 1.05 - 0.03\{0.6 + 4 \log_{10}\}$

$= 1.05 - 0.03\{4.6\} = 0.912$ V

BIOLOGY

51) NCERT, Pg # E-26, 27, 29, 30, H-29, 29, 31, 32

52)

NCERT, Pg # E-44, 45, 46, H-49, 50, 51

53) NCERT-,Pg # H-28, E-26

54) NCERT-, E-23, H-30

55) NCERT, Pg # 28, H-30

56) NCERT, Pg # E-31, H-33

57) NCERT, Pg # 32, H-35

58) NCERT, E-32, H-35

59) NCERT, Pg # E-32, H-36

60) NCERT, Pg # Module

61) NCERT, Pg # Module

62) NCERT, Pg # E-36, H-39

63) NCERT, Pg # E-41, H-46, 47

64) NCERT, Pg # 3

65) NCERT, Pg # E-38, H-42

66) NCERT, Pg. # 37

67) NCERT, Pg. # 36

68) NCERT P.No. 29, 30

69) NCERT-XII, Pg. # 33, Para-2.4

70) NCERT Pg # 31

71) NCERT, Pg # 29

72) NCERT Pg. # 27

73) NCERT XII Pg # 26 (E), 28 (H)

74) Bartholin glands ⊕nt in female not male
बार्थोलिन ग्रन्थि मादा में उपस्थित होती है न कि नर मे

75) Inhibin secret by granulose cells & inhibits the secretion of FSH.
इन्हिबिन कणिका कोशिका के द्वारा स्त्रावित होता है जो FSH के स्त्रावण को संदमित करता है।

76)

NCERT (XII) Pg. No. # 27

77) NCERT Pg. # 49

78)

Colostrum is nutritious contains proteins, several antibody, essential fatly acids for proper development of baby.

79)

NCERT-XII, Pg. # 34

80)

NCERT-XII, Pg. # 27

81)

NCERT XII, Pg. # 46

82)

NCERT-XII, Pg. # 29

83)

NCERT Pg. 43

84)

NCERT-XII, Pg. # 33

85) NCERT XII Pg. # 47 (E) 51 (H)

86) NCERT, Pg. # 38

87) NCERT, Pg. # 34
During proliferative phase, endometrium proliferates under the effect of estrogen hormone which is secreted by developing ovarian follicles.

88) NCERT P.No. Module

89) NCERT P.No. 26

90) NCERT P. No. 32

91) NCERT, Pg # 34,37,38

92)

NCERT Pg.No-36

93)

NCERT Pg # 50

94) NCERT Pg. # 47

95) NCERT, Pg. # 48

96) NCERT XII Pg. No. # 42

97) NCERT Pg # 44-45 (E)
Pg # 49-50 (E)

98) NCERT Pg # 42

99) NCERT, Pg. # 44 (E)

100) NCERT (XII) Pg. # 63, Para-4.4